BELLABEAT

CUSTOMER'S SMART DEVICE USAGE, SLEEP AND ACTIVITY PATTERNS

CREATED BY: JAN PRUSZCZYŃSKI

CREATED ON: 11.10.2026

DATA FROM: 12.03.2016 - 12.05.2016

AGENDA

EXECUTIVE SUMMARY

METHODOLOGY AND LIMITATIONS

DAILY ACTIVITY USER DATA

HOURLY STEPS/INTENSITY AND SLEEPING HABITS

RECOMMENDATIONS

EXECUTIVE SUMMARY

OBJECTIVE: UNDERSTAND HOW SMART DEVICE USERS MOVE, REST AND TRACK SLEEP, TO IDENTIFY TRENDS THAT COULD BE RELEVANT TO **BELLABEAT'S LEAF USERS** AND **RECOMMEND MARKETING STRATEGIES.**

DATA: DAILY AND HOURLY ACTIVITY AND SLEEP RECORDS **FROM 35 USERS OVER TWO MONTHS.**

KEY FINDINGS:

- **SATURDAY IS THE MOST ACTIVE DAY** (7,752 STEPS) AND **SUNDAY THE LOWEST** (6,607), AND ALL DAYS STAY BELOW 10,000 STEPS.
- **MOST ACTIVITY IS LIGHT** (ABOUT 190 MIN A DAY) **VERSUS ABOUT 20 VERY ACTIVE MINUTES**, AND SEDENTARY TIME RUNS 16 TO 17 HOURS.
- **ACTIVITY PEAKS AROUND NOON AND BETWEEN 5 AND 7 PM, THEN DROPS AFTER 8 PM.**
- ALL 35 USERS TRACKED ACTIVITY DAILY, BUT ONLY ABOUT 16 TO 21 LOGGED SLEEP ON ANY GIVEN DAY.

RECOMMENDATIONS: SEND PERSONALISED EXERCISE NUDGES BETWEEN 8 AM AND 7 PM, PROMPT SLEEP LOGGING AT 8 TO 9 PM, AND TAILOR MESSAGES TO THE DAY OF THE WEEK.

METHODOLOGY AND LIMITATIONS

METHODOLOGY

DATA SOURCE: PUBLIC FITBIT FITNESS TRACKER DATASET (KAGGLE), WITH DAILY ACTIVITY, HOURLY STEPS AND INTENSITY, AND SLEEP RECORDS FROM 35 USERS OVER TWO MONTHS.

TOOLS: BIGQUERY (SQL) FOR CLEANING AND AGGREGATION, **TABLEAU PUBLIC** FOR THE CHARTS, **CANVA** FOR THE PRESENTATION.

PREPARATION: CHECKED THE TABLES FOR DUPLICATES AND MISSING VALUES, MATCHED USERS ACROSS TABLES BY ID, AND AVERAGED EACH MEASURE BY DAY OF THE WEEK AND BY HOUR OF THE DAY.

ANALYSIS: COMPARED AVERAGE STEPS, ACTIVITY MINUTES (VERY, FAIRLY, LIGHTLY ACTIVE AND SEDENTARY), HOURLY PATTERNS, AND THE NUMBER OF UNIQUE USERS WHO LOGGED SLEEP EACH DAY.

LIMITATIONS

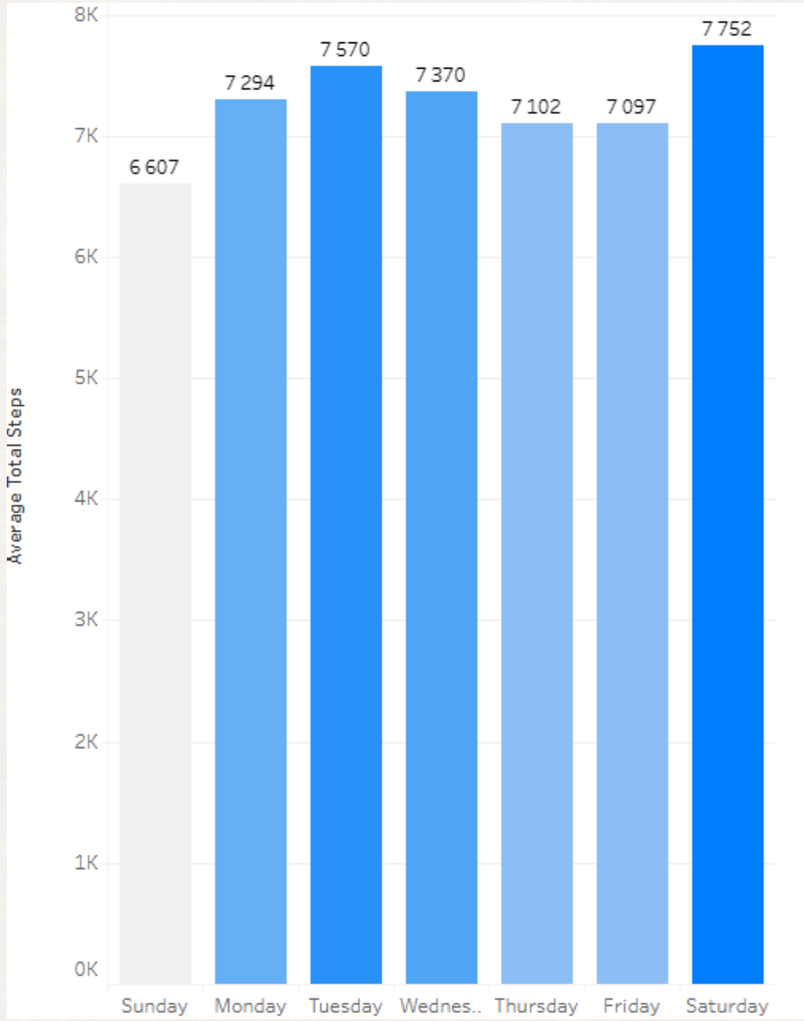
- **SMALL SAMPLE OF 35 USERS**, WITH NO DEMOGRAPHIC INFORMATION (AGE, GENDER).
- **DATA FROM 2016,** SO HABITS AND DEVICES MAY HAVE CHANGED**.**
- **SHORT PERIOD,** ONLY TWO MONTHS WORTH OF DATA RANGING FROM 12.03.2016 – 12.05.2016.
- **SLEEP DATA COVERS ONLY ABOUT 16 TO 21 USERS PER DAY,** SO SLEEP AVERAGES ARE LESS RELIABLE.
- **AVERAGES DESCRIBE THE GROUP AS A WHOLE AND SAY NOTHING ABOUT INDIVIDUAL USERS.**

PART 1: DAILY ACTIVITY USER DATA

AVERAGE DAILY STEPS

- STEPS PEAK ON **SATURDAY** (7,752) AND **TUESDAY** (7,570).
- **SUNDAY IS THE LEAST ACTIVE DAY** (6,607), ROUGHLY **15% BELOW** SATURDAY.
- AVERAGE WEEKDAY STEPS (7,287) ARE **HIGHER** THAN AVERAGE WEEKEND STEPS (7,188)

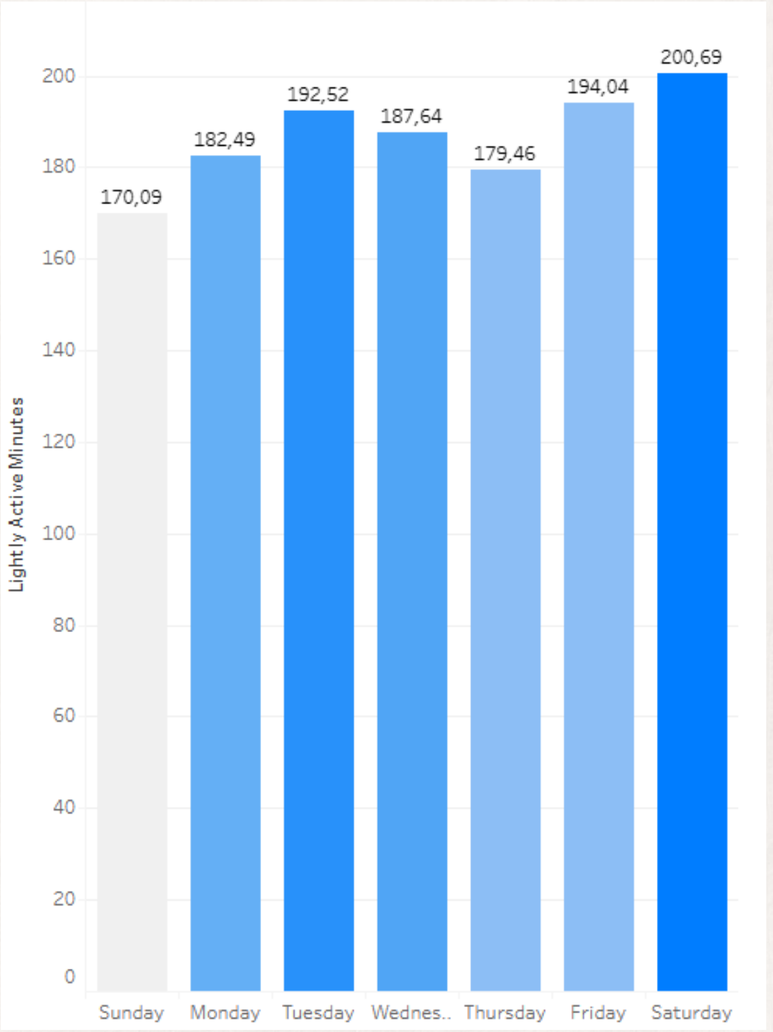
(DRIVEN BY SUNDAY'S LOW)



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LIGHTLY ACTIVE MINUTES

- LIGHTLY ACTIVE TIME **PEAKS ON SATURDAY** (200.69 MIN) AND **FRIDAY** (194.04 MIN).
- **SUNDAY IS THE LOWEST DAY** (170.09 MIN), ABOUT **15% BELOW SATURDAY.**
- OVERALL ACTIVITY IS STEADY, STAYING **BETWEEN 170 AND 200 MINUTES DAILY.**



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FAIRLY VS VERY ACTIVE MINUTES

- VERY ACTIVE TIME IS **HIGHEST ON SATURDAY** (20.81 MIN) AND **LOWEST ON THURSDAY** (17.06 MIN).
- **FAIRLY ACTIVE TIME PEAKS ON FRIDAY** (15.76 MIN) AND STAYS NEAR 12 TO 13 MINUTES ON MOST OTHER DAYS.

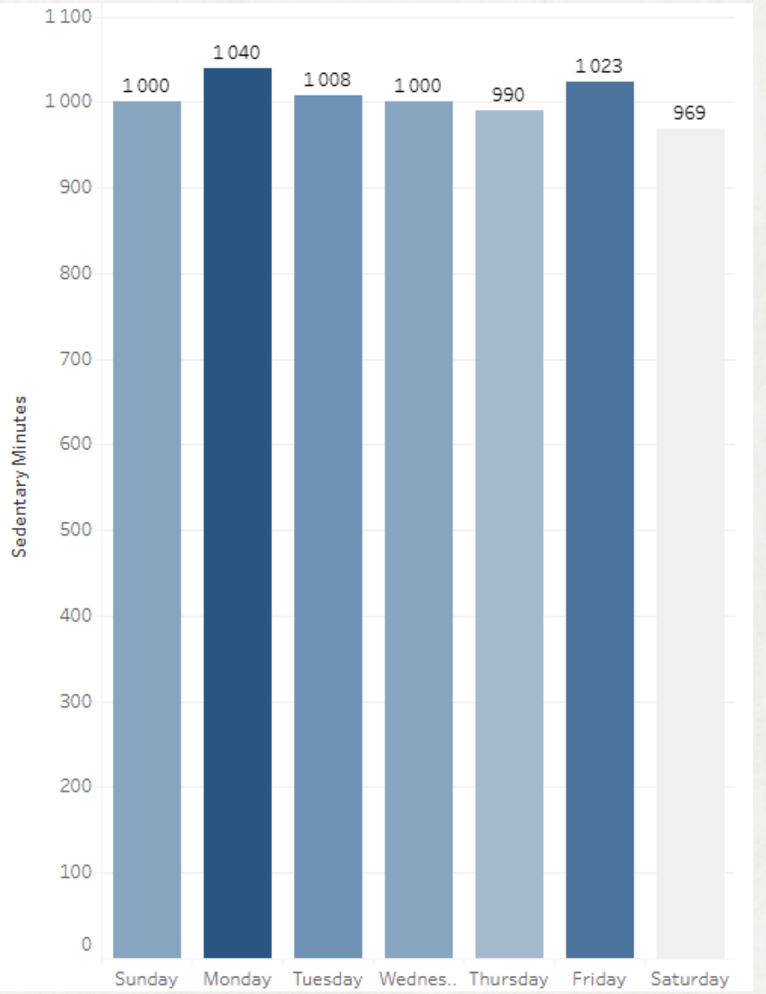


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SEDENTARY MINUTES

- SEDENTARY TIME STAYS B**ETWEEN 969 AND 1,040 MINUTES A DAY**, ROUGHLY 16 TO 17 HOURS.

- **IT PEAKS ON MONDAY** (1,040 MIN) AND DROPS TO I**TS LOWEST ON SATURDAY** (969 MIN), THE MOST ACTIVE DAY.



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DAILY ACTIVITY SUMMARY

- **Saturday is the most active day.** It leads in steps (7,752), very active minutes (20.81) and lightly active minutes (200.69), and has the least sedentary time (969 min).

- **Sunday is the quietest day for movement.** It has the fewest steps (6,607) and the fewest lightly active minutes (170.09).

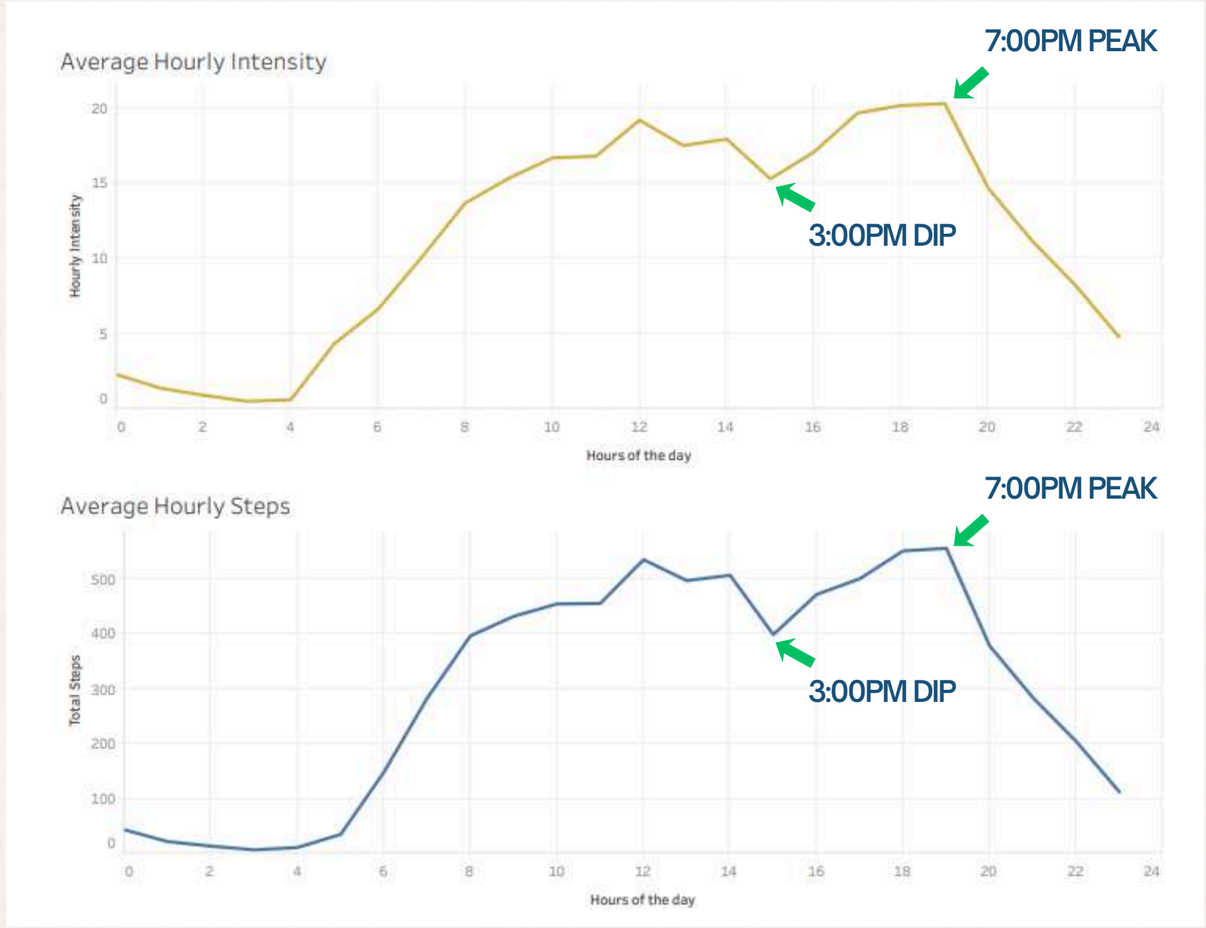
- **Activity is mostly low intensity.** Users log about 190 lightly active minutes a day versus about 20 very active and about 13 fairly active.

- **Sedentary time dominates.** At 969 to 1,040 minutes a day (roughly 16 to 17 hours), it barely changes through the week.

PART 2: HOURLY STEPS/INTENSITY AND SLEEPING HABITS

AVERAGE HOURLY STEPS AND INTENSITY

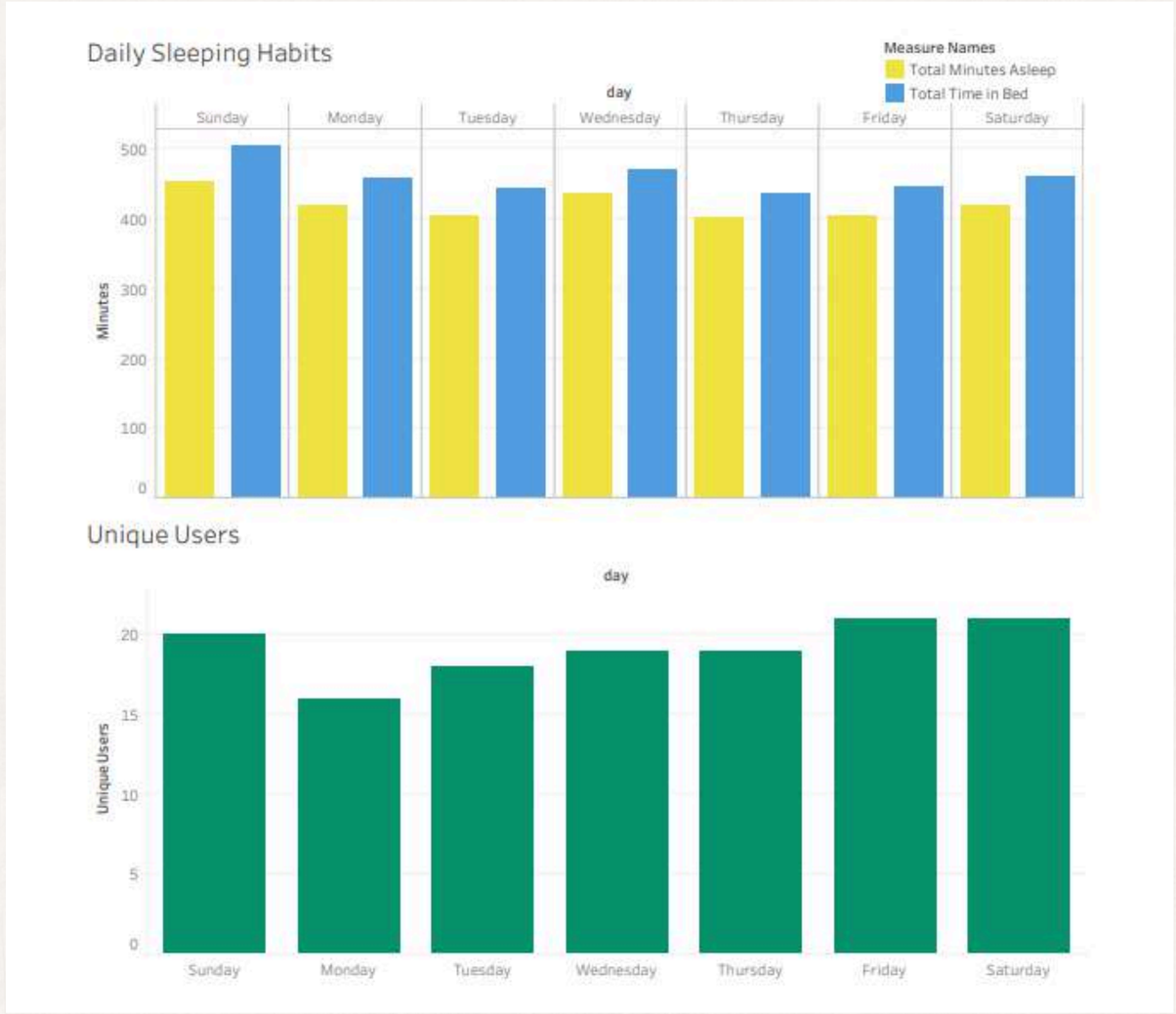
- **ACTIVITY IS LOWEST OVERNIGHT** AND RISES SHARPLY FROM 5 TO 6 A.M.
- **TWO PEAKS:** AROUND NOON AND BETWEEN 5 AND 7 P.M., WITH THE EVENING PEAK THE HIGHEST.
- **BOTH STEPS AND INTENSITY FALL QUICKLY AFTER 8 P.M.**



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DAILY SLEEPING HABITS

- **ALL 35 USERS LOGGED ACTIVITY DAILY,** BUT **ONLY ABOUT 16 TO 21 LOGGED SLEEP** ON ANY GIVEN DAY.
- SLEEP LOGGING **PEAKS ON FRIDAY AND SATURDAY** AND IS **LOWEST ON MONDAY.**
- **USERS SLEEP ABOUT 7 TO 8 HOURS ON AVERAGE**, WITH TIME IN BED SLIGHTLY HIGHER.



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HOURLY ACTIVITY AND SLEEP HABITS SUMMARY

- **Activity follows a clear daily rhythm.** Steps and intensity are lowest overnight, climb sharply from 5 to 6 a.m., and peak twice: around noon and between 5 and 7 p.m. **The evening peak is the highest.**

- **Activity falls quickly after 8 p.m.** Both measures drop steeply into the night, with a small dip around 3 p.m.

- **Sleep tracking is far less consistent than activity tracking.** All 35 users logged activity every day, but only about 16 to 21 logged sleep on any given day, and fewer than half did on Mondays.

- **Users who do log sleep get about 7 to 8 hours.** Time in bed runs slightly above time asleep, but with so few records these averages should be read with caution.

PART 3: RECOMMENDATIONS

RECOMMENDATIONS

01

SEND EXERCISE NUDGES BETWEEN 8 AM AND 7 PM.

TIME THEM AROUND THE 3 PM DIP AND THE EVENING PEAK.

02

PERSONALISE BY ACTIVITY LEVEL.

LIGHTLY ACTIVE MINUTES MAKE UP MOST LOGGED ACTIVITY, SO ENCOURAGE THESE USERS TO MOVE MORE AND GIVE VERY ACTIVE USERS CHALLENGES THAT FIT THEIR LEVEL.

03

PROMPT SLEEP TRACKING AT 8 TO 9 PM.

REMIND USERS TO WEAR THEIR DEVICE OVERNIGHT SO SLEEP DATA BECOMES AS COMPLETE AS ACTIVITY DATA.

04

TAILOR MESSAGES TO THE DAY.

PUSH EXERCISE ON SATURDAY AND MIDWEEK LOWS (THURSDAY, FRIDAY), AND SEND LIGHT MOVEMENT OR RECOVERY CONTENT ON SUNDAY.

SUCCESS METRICS

RECOMMENDATIONS	METRIC	HOW TO MEASURE
Send exercise nudges (8AM to 7PM)	Average daily steps per user	Compare users who get nudges with a group who don’t, or compare before and after launch
Personalise by activity level	Share of lightly active users who move up a level	Track lightly active users who add very or fairly active minutes over time
Sleep logging prompt (8 to 9 PM)	Share of users logging sleep each day	Currently about 16 to 21 of 35 (roughly 45% to 60%), so set a target above that
Day-specific messages	Steps on the weakest days	Watch Sunday (6,607 steps) and Thursday/Friday (about 7,100) against the weekly average

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THANK YOU FOR LISTENING

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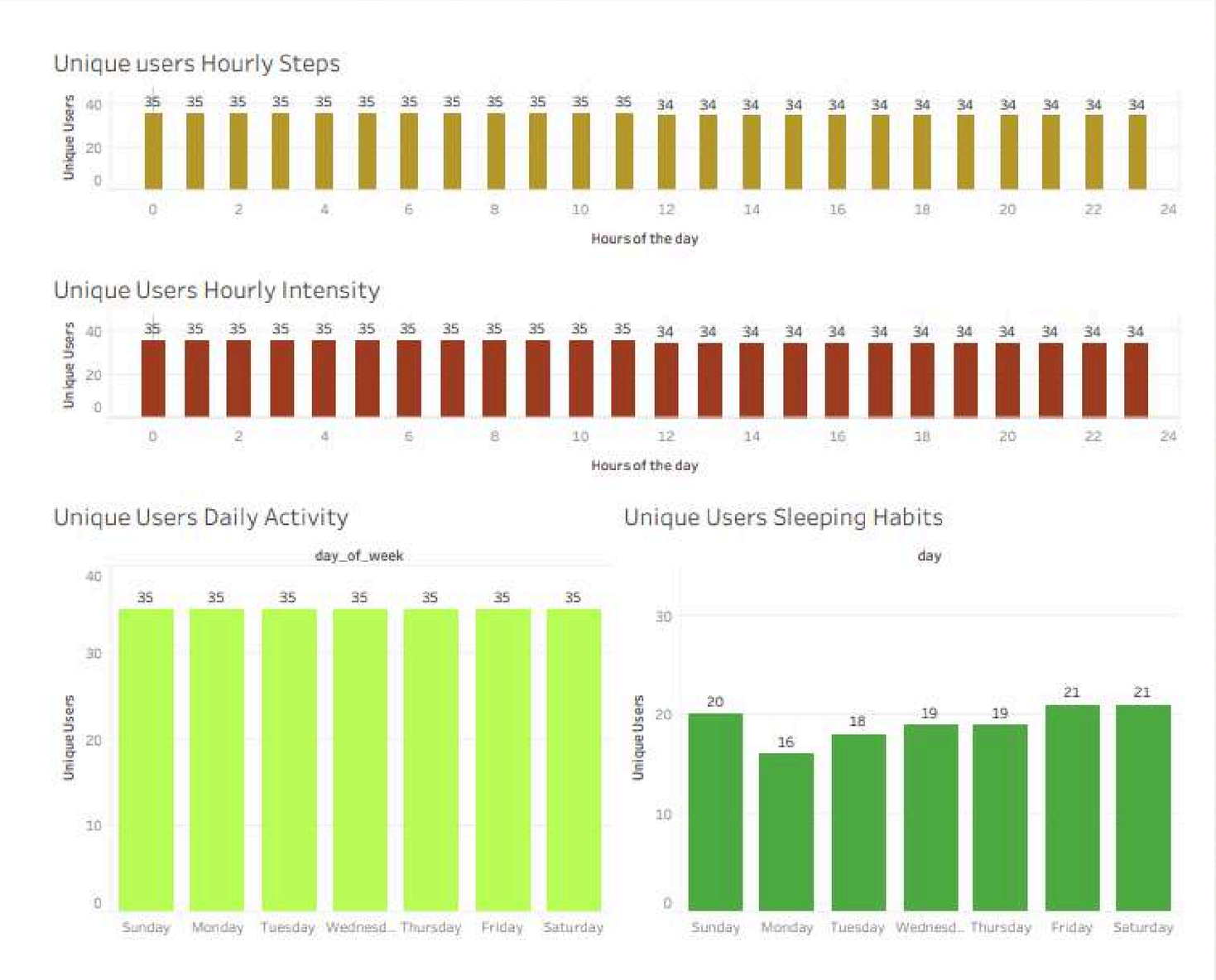
APPENDIX

DAILY ACTIVITY USER DATA DASHBOARD



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UNIQUE USERS DAILY/HOURLY/SLEEPING HABITS



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